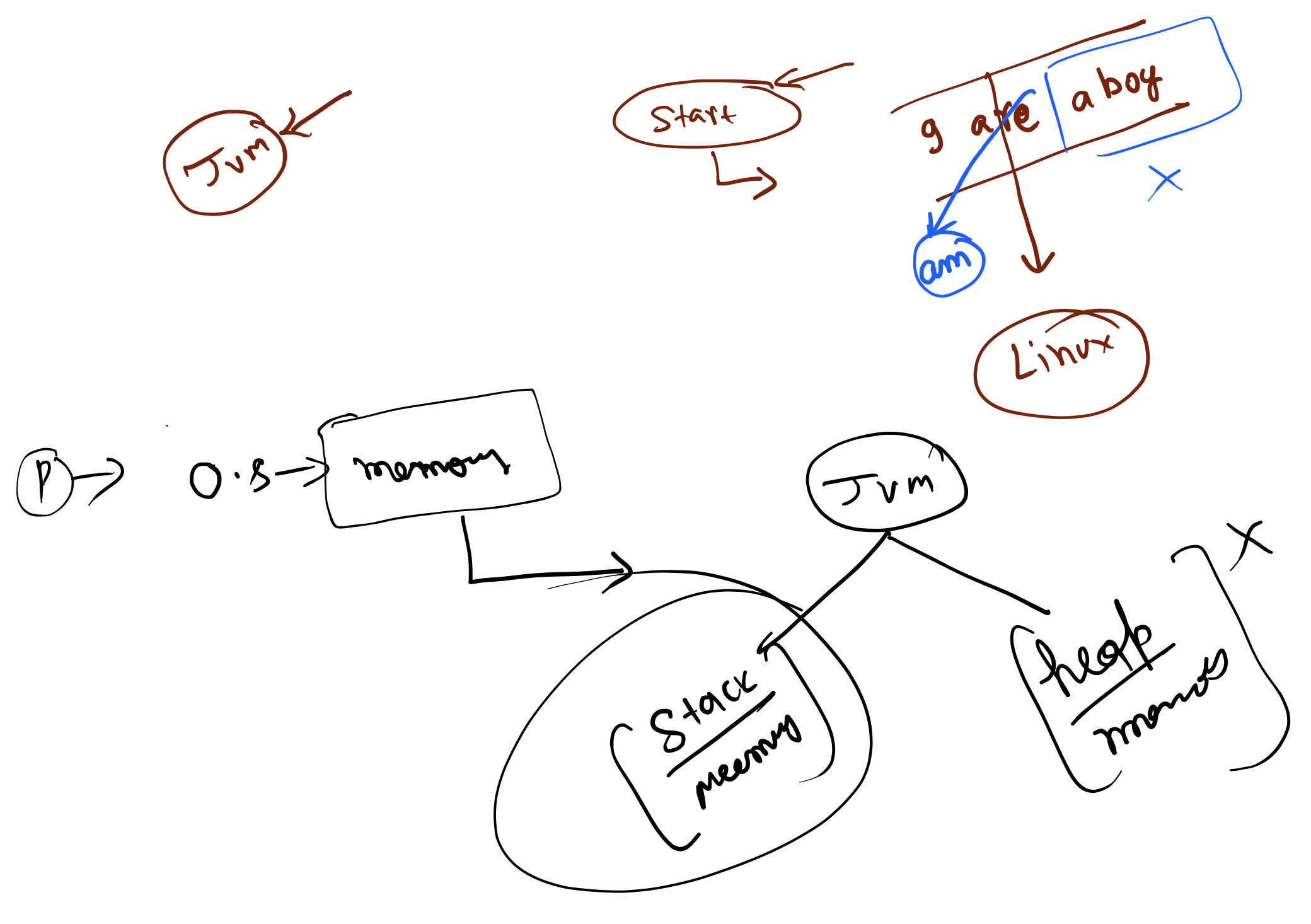
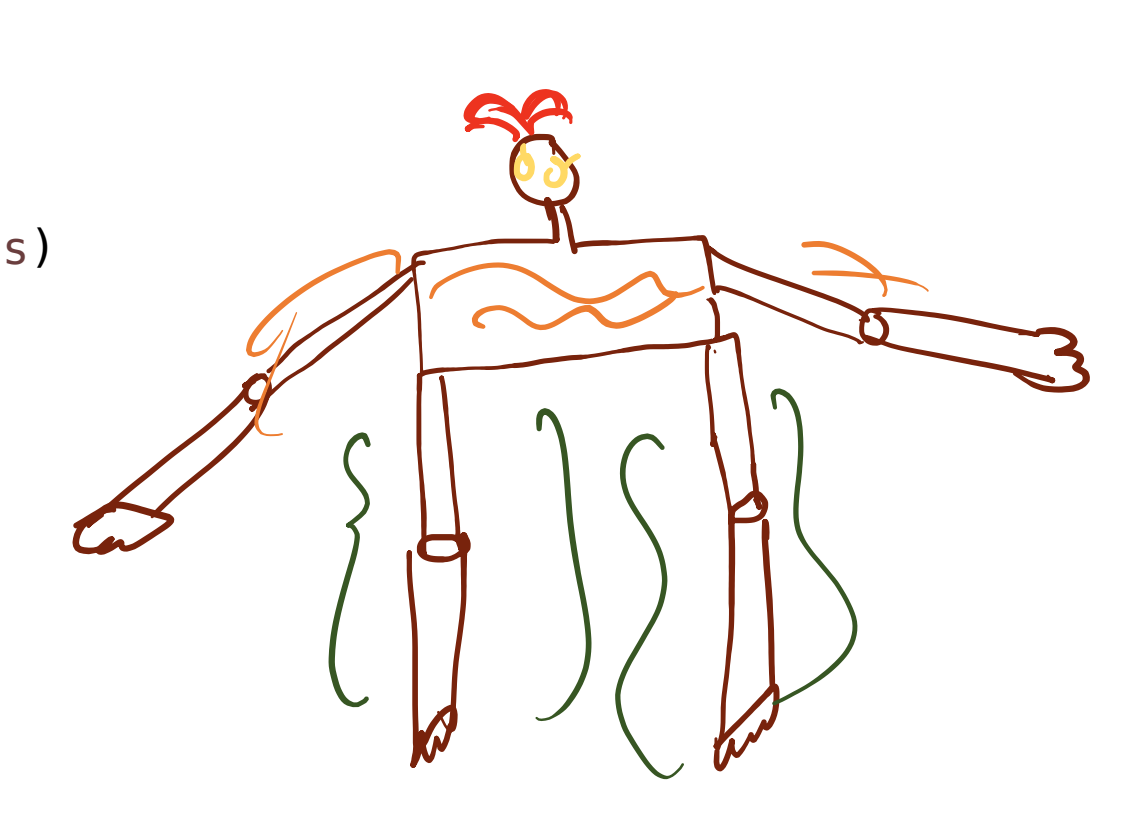
Lec-5

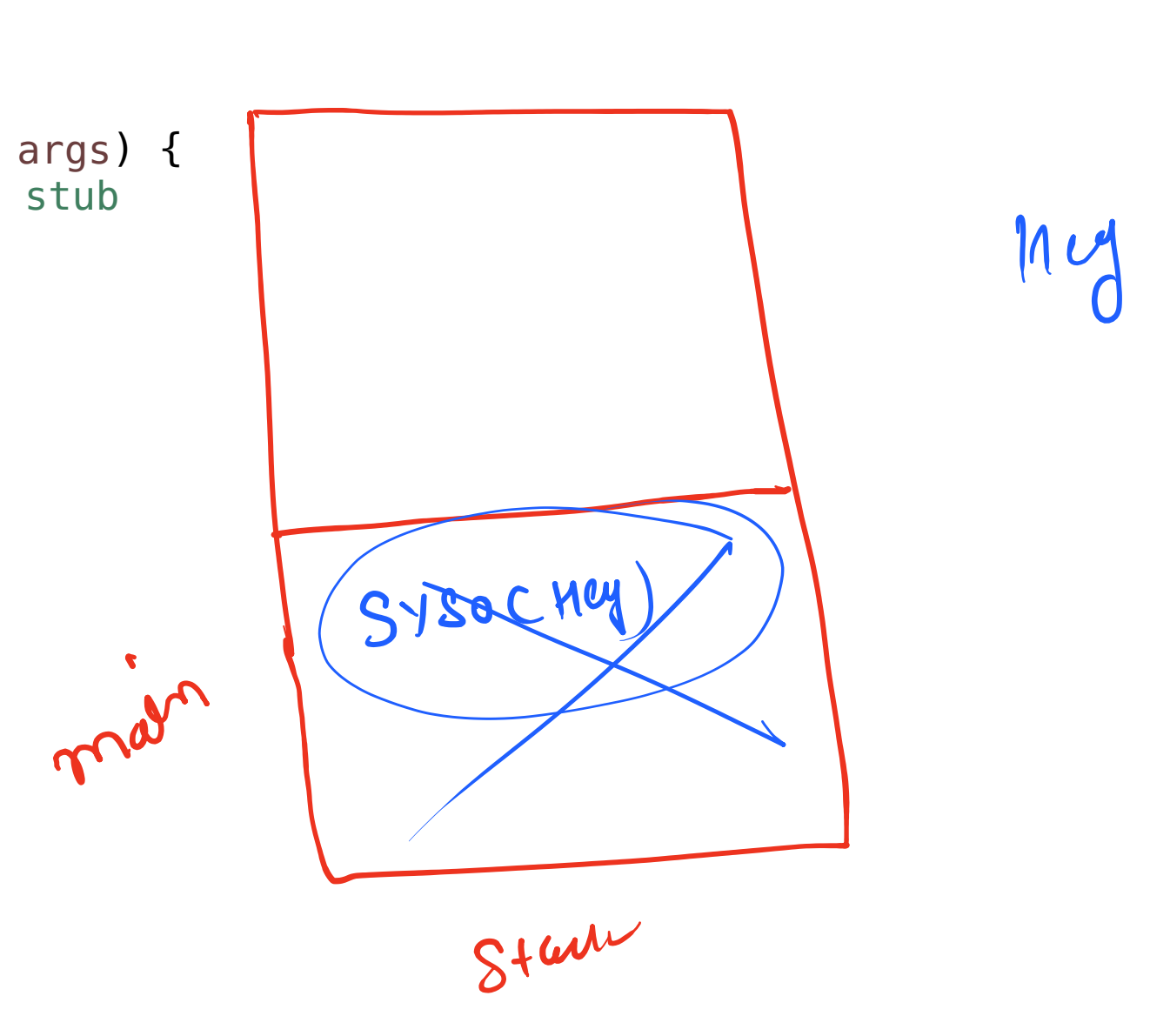
06/06/26 12:59 PM

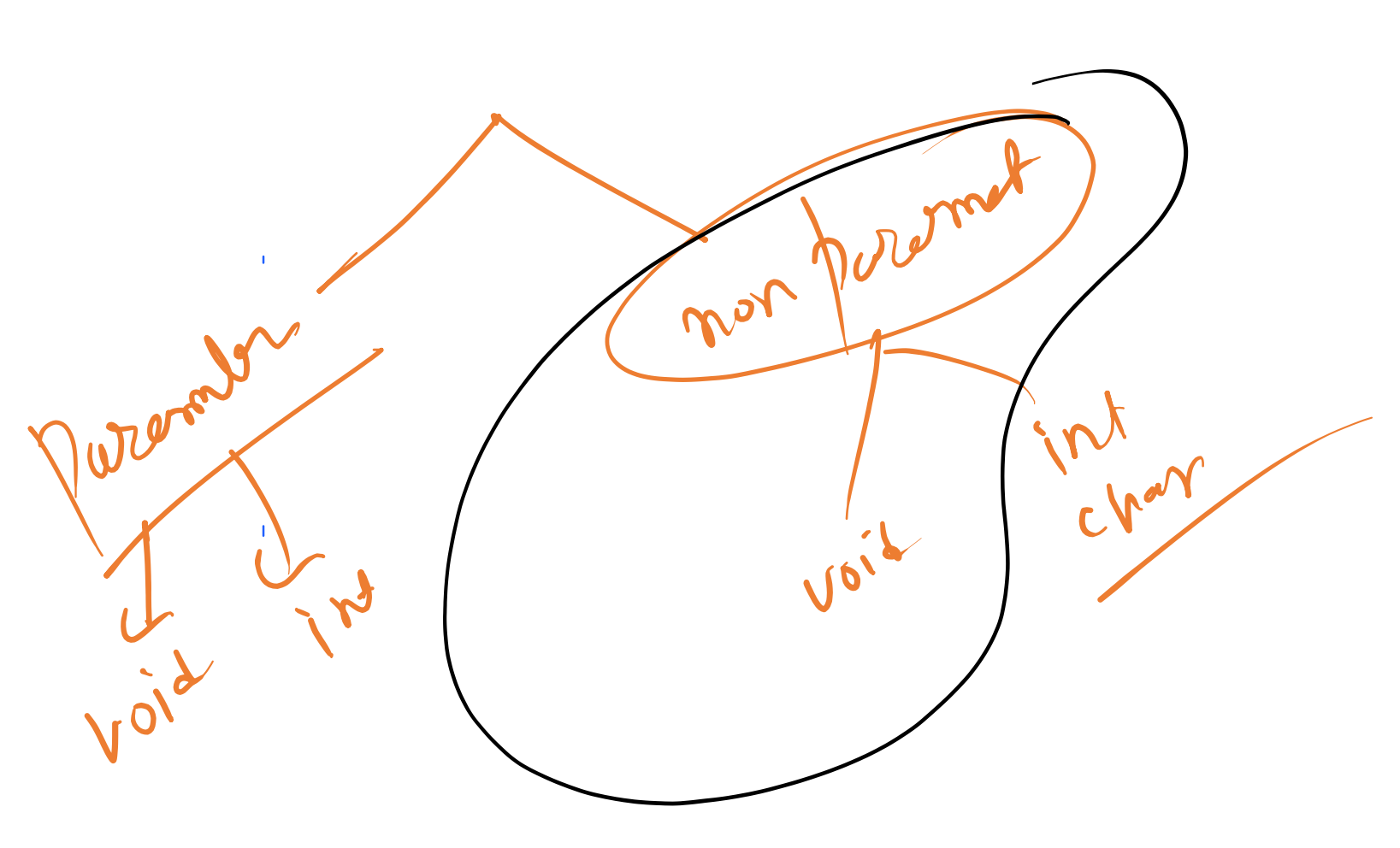

Jvm
Start
I ate a boy
am
Linux
P
O.S
memory
Jvm
Stack memory
heap memory

```
public static void main(String[] args)
```

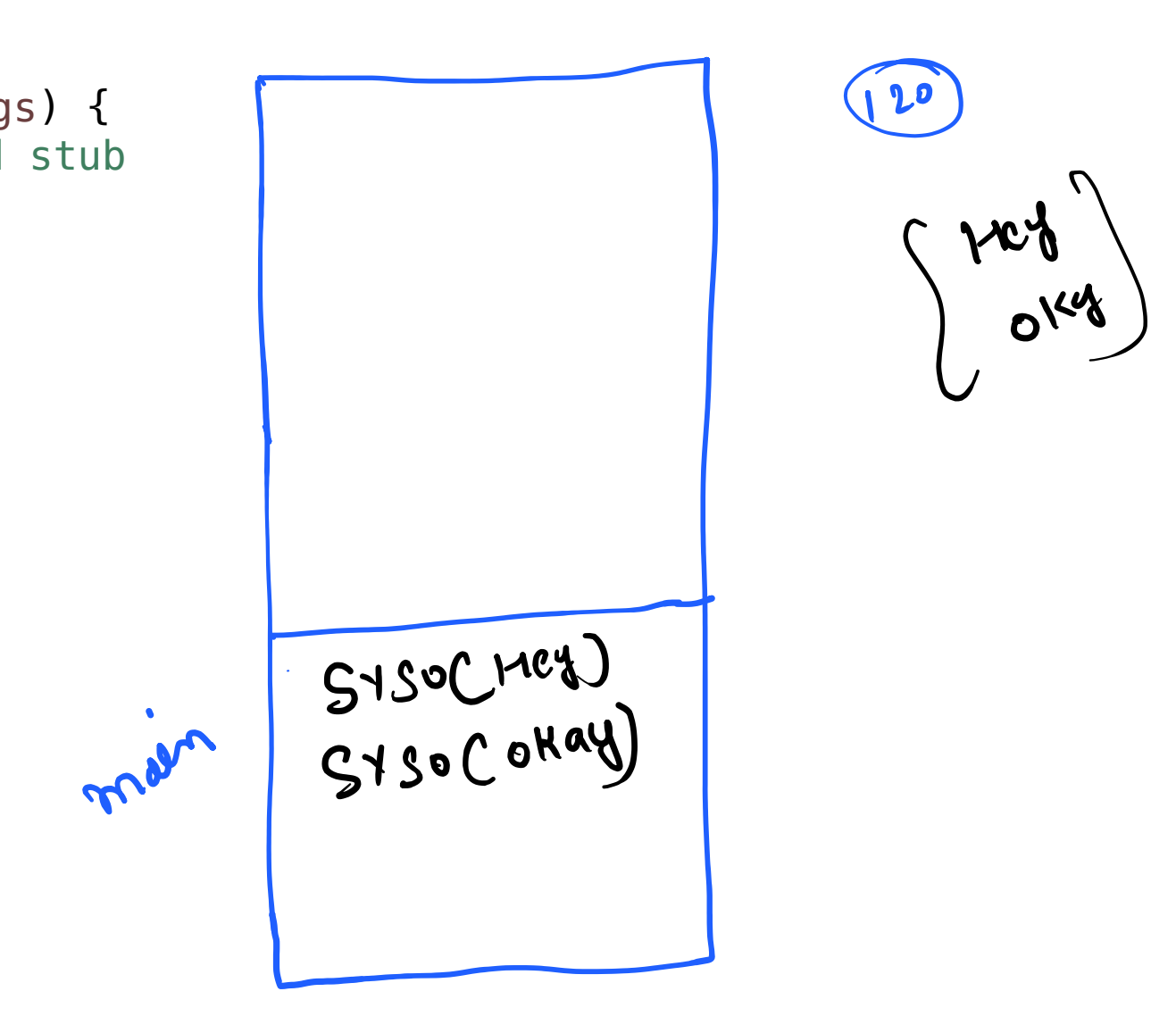


```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");

    }
```

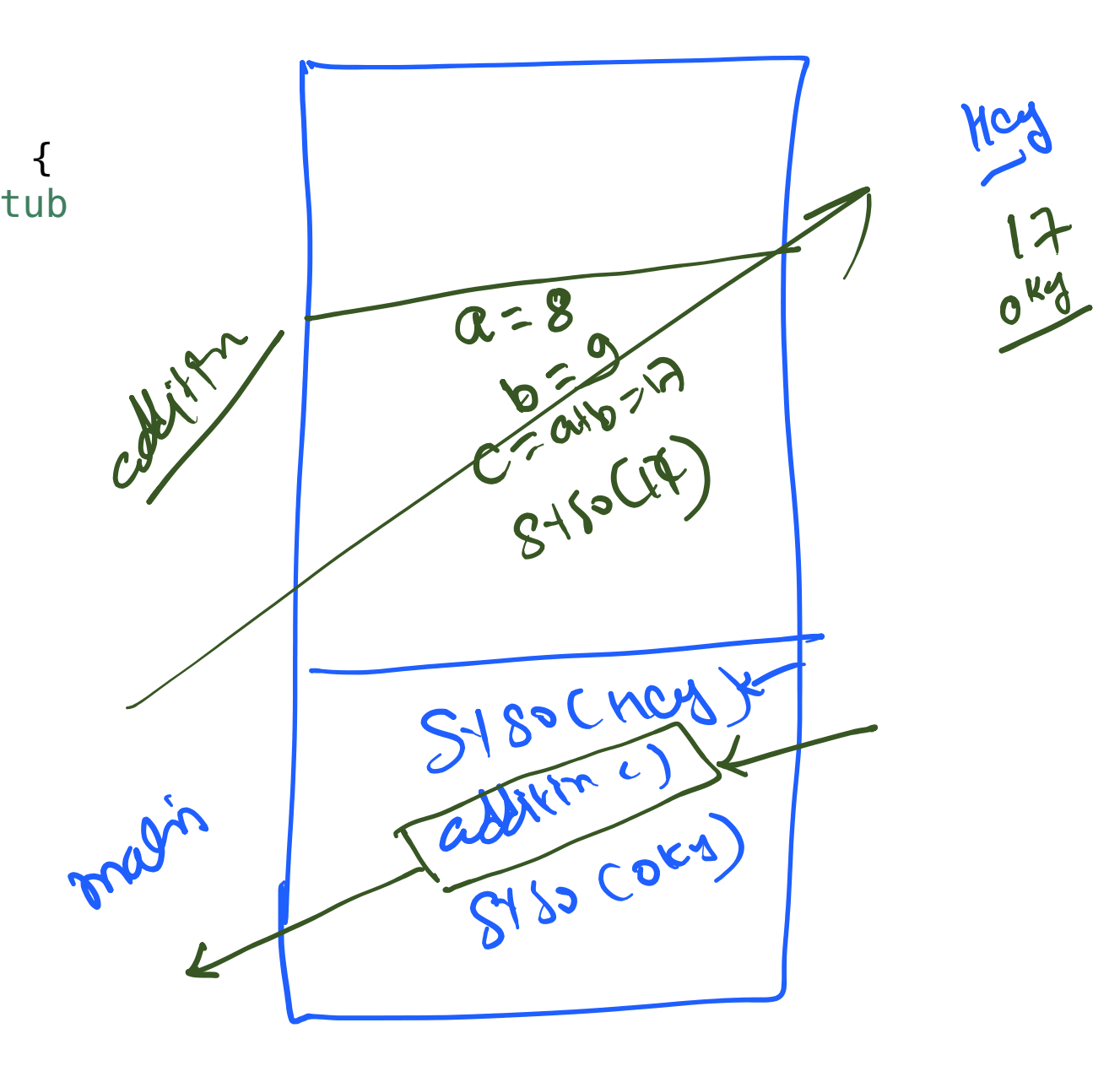

Sysoc(Hey)
main
Stack
Hey


Paramter
void
int
non paramet
void
int
char

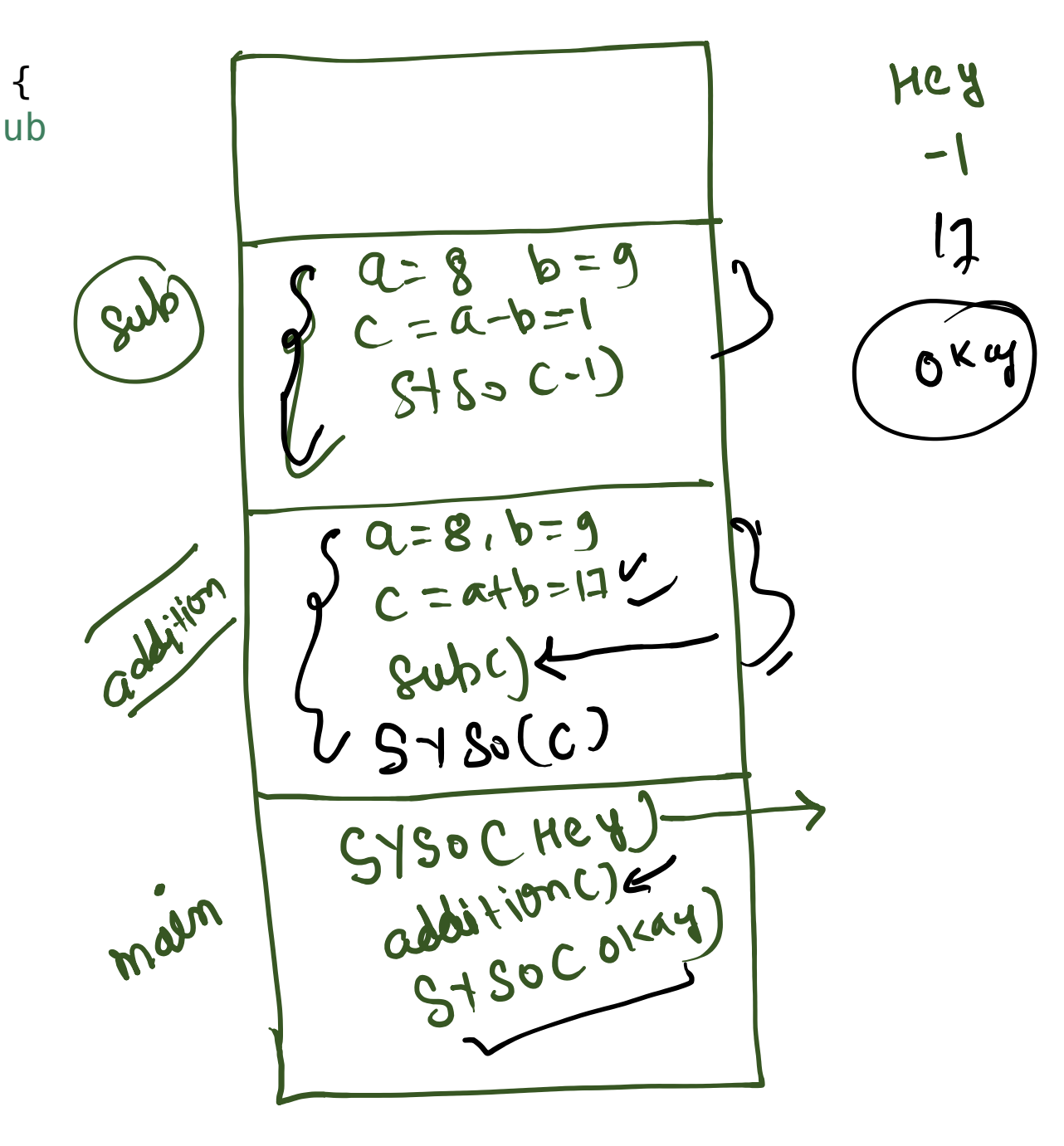
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        System.out.println("Okay");

    }
    public static void addition() {
        int a = 8;
        int b = 9;
        int c = a + b;
        System.out.println(c);
    }
```


Syso(Hey)
Syso(Okay)
main
120
Hey
okay

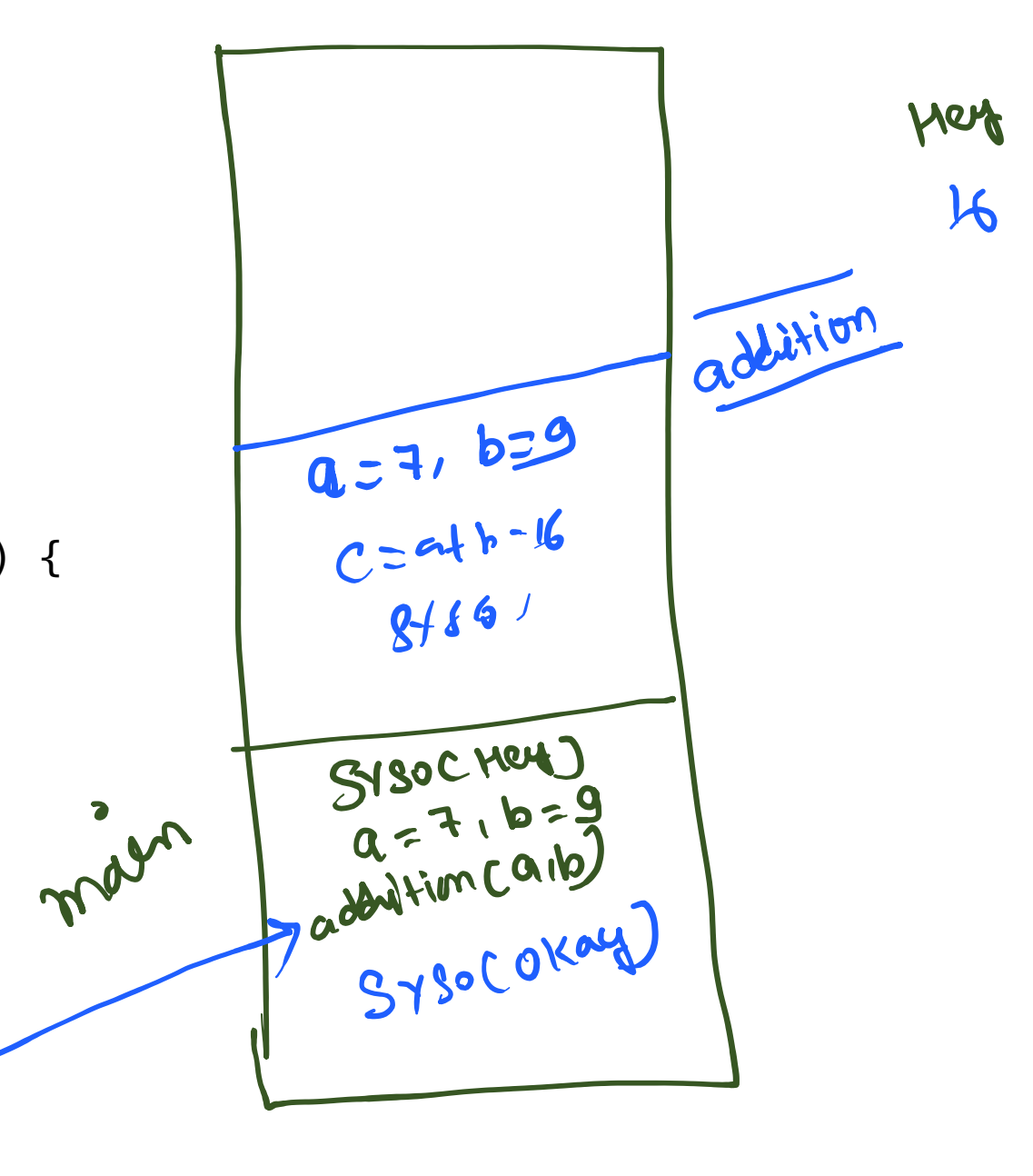
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        addition();
        System.out.println("Okay");

    }
    public static void addition() {
        int a = 8;
        int b = 9;
        int c = a + b;
        System.out.println(c);
    }
```


addition
a=8
b=9
c=a+b=17
Syso(17)
Syso(Hey)
addition()
Syso(Okay)
main
Hey
17
okay

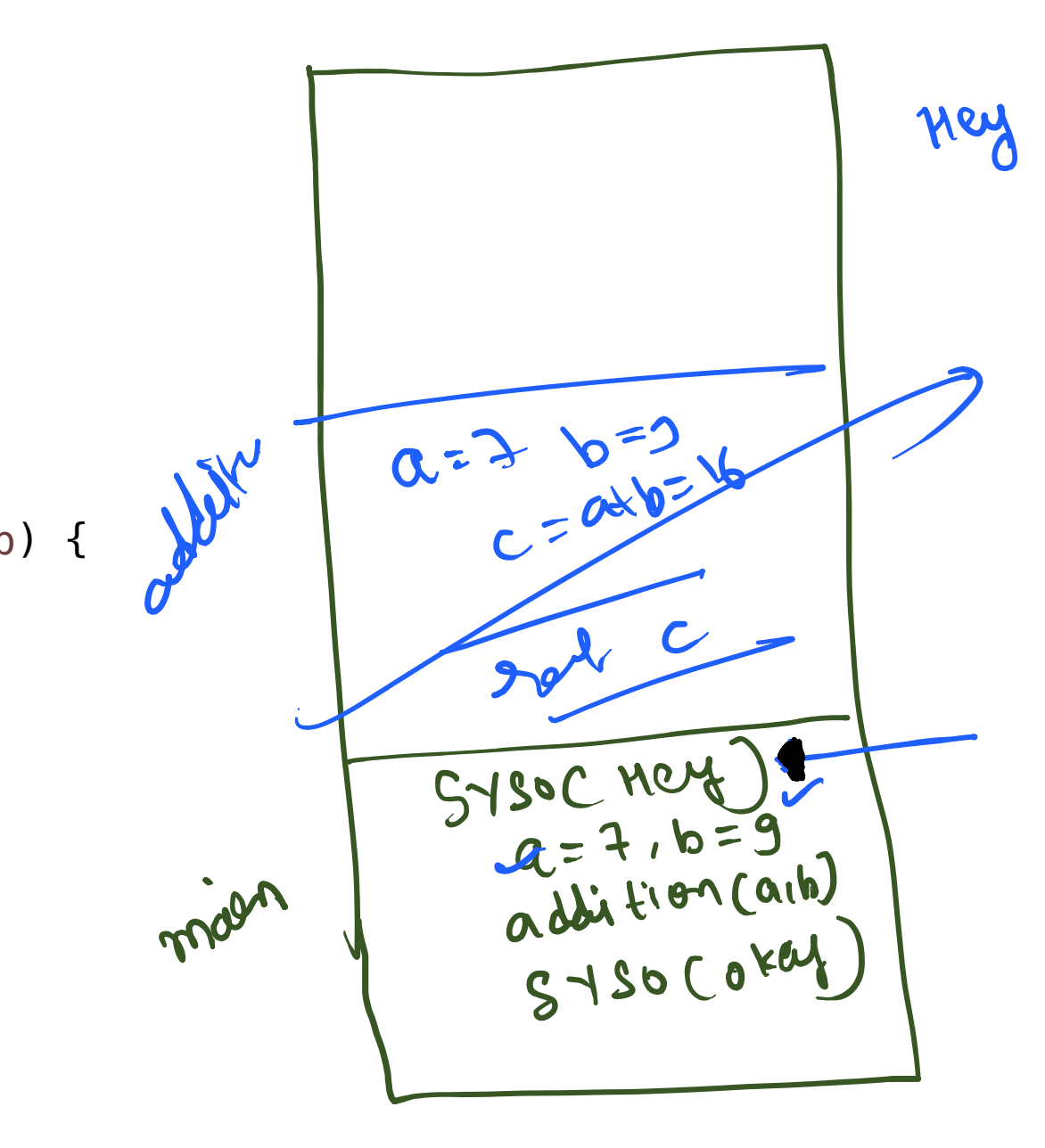
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        addition();
        System.out.println("Okay");

    }
    public static void addition() {
        int a = 8;
        int b = 9;
        int c = a + b;
        sub();
        System.out.println(c);
    }
    public static void sub() {
        int a = 8;
        int b = 9;
        int c = a - b;
        System.out.println(c);
    }
```


sub
a=8 b=9
c=a-b=-1
Syso(-1)
addition
a=8, b=9
c=a+b=17
sub()
Syso(c)
main
Syso(Hey)
addition()
Syso(okay)
Hey
-1
17
okay

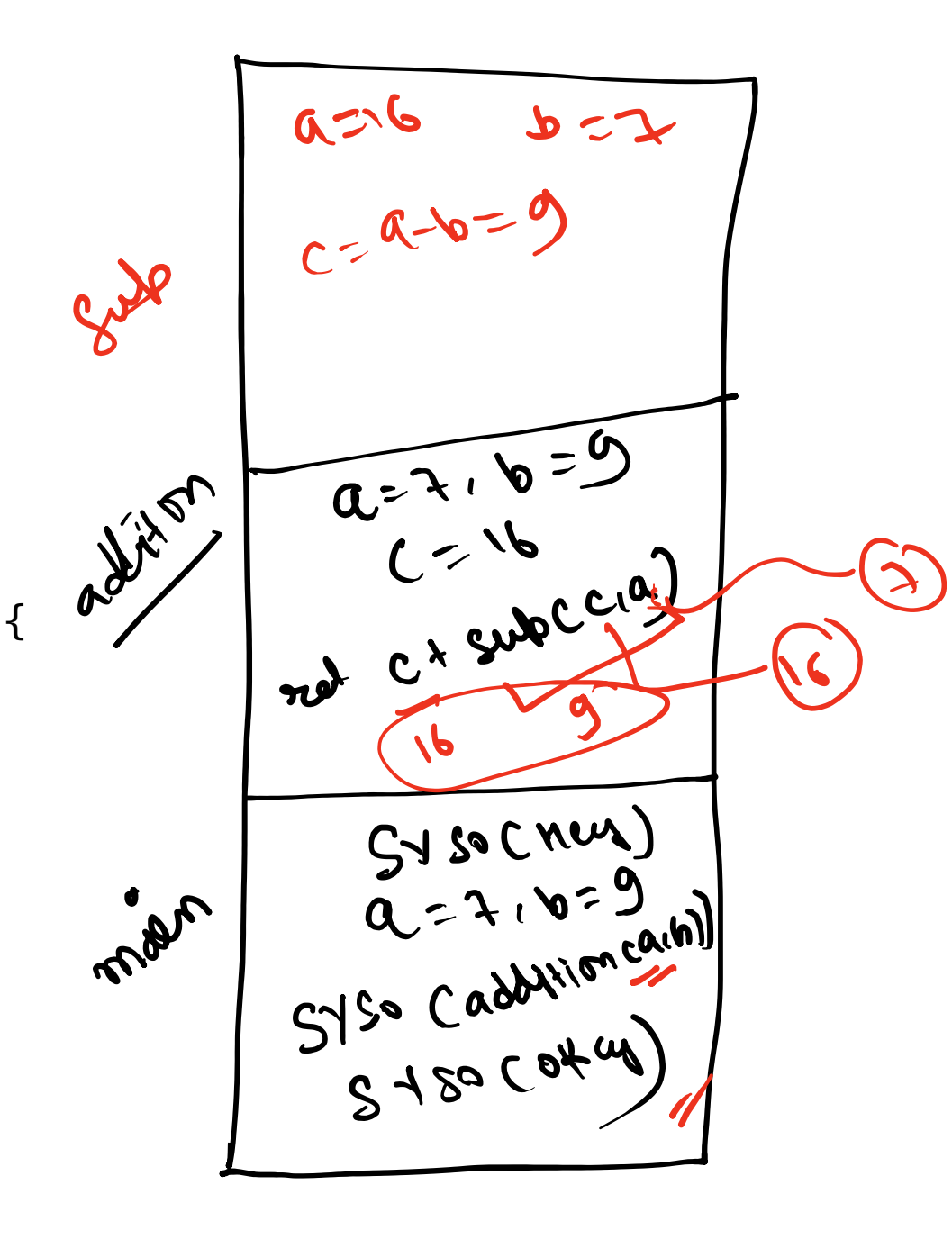
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        int a = 7;
        int b = 9;
        addition(a, b);
        System.out.println("Okay");

    }
    public static void addition(int a, int b) {

        int c = a + b;
        System.out.println(c);
    }
```


Hey
16
addition
a=7, b=9
c=a+b=16
Syso(
main
Syso(Hey)
a=7, b=9
addition(a,b)
Syso(okay)

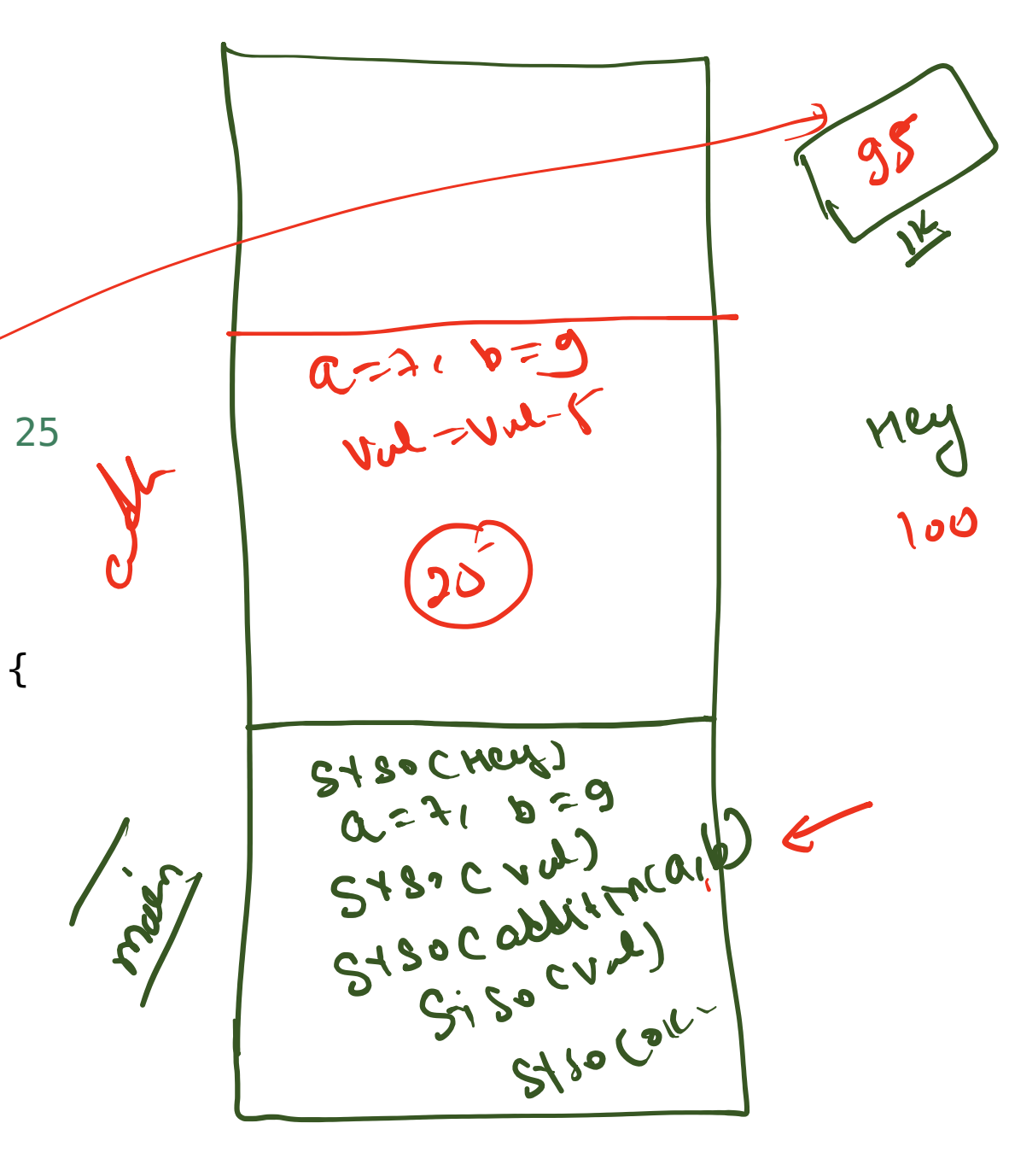
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        int a = 7;
        int b = 9;
        addition(a, b);
        System.out.println("Okay");

    }
    public static int addition(int a, int b) {

        int c = a + b;
        return c;
    }
```


7,9
16
Hey
addition
a=7 b=9
c=a+b=16
ret c
main
Syso(Hey)
a=7, b=9
addition(a,b)
Syso(okay)

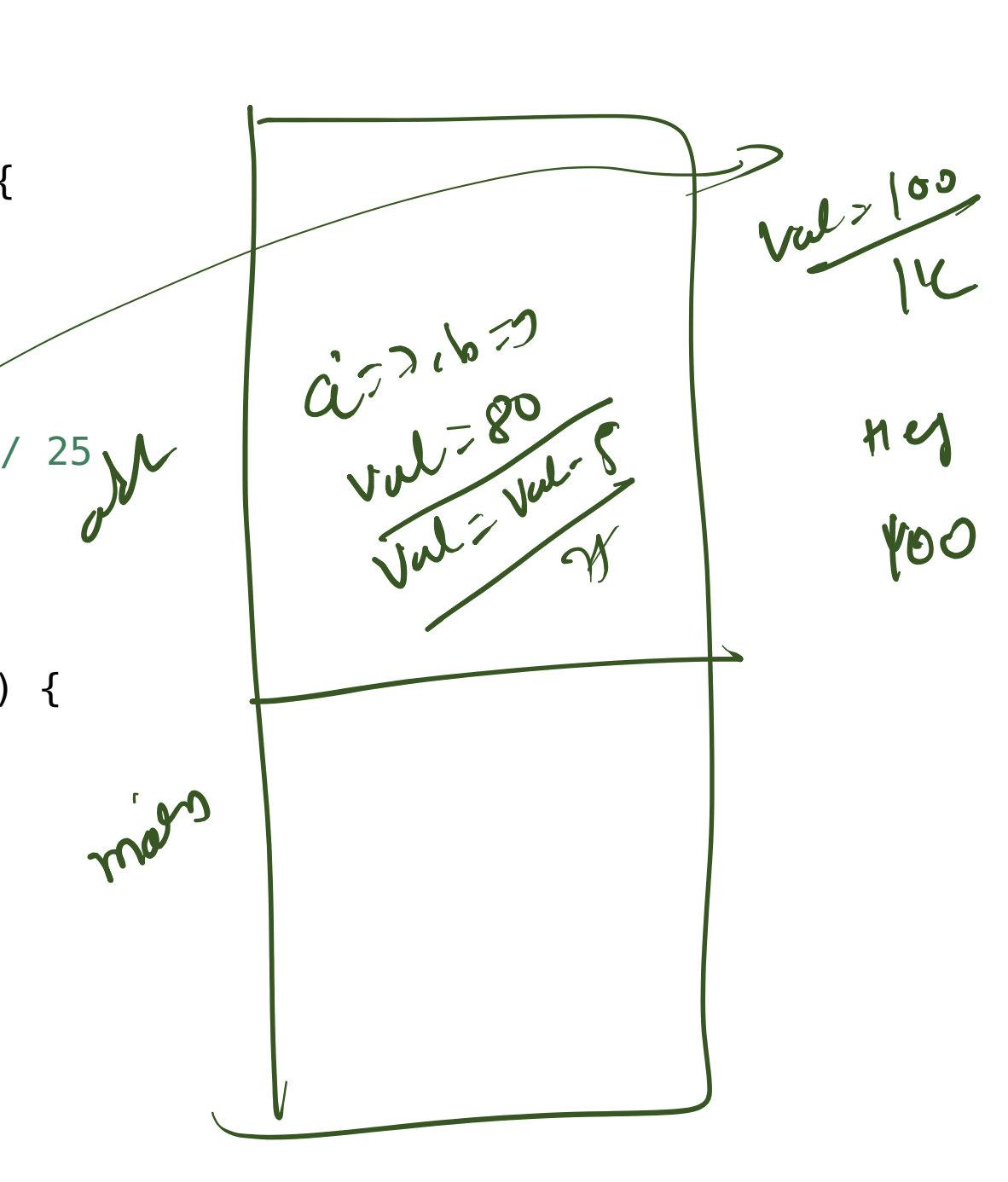
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        int a = 7;
        int b = 9;
//      int ans=addition(a, b);
//      System.out.println(ans);
        System.out.println(addition(a, b));
        System.out.println("Okay");

    }
    public static int addition(int a, int b) {

        int c = a + b;
        return c + sub(c, a);
    }
    public static int sub(int a, int b) {
        return a - b;
    }
```

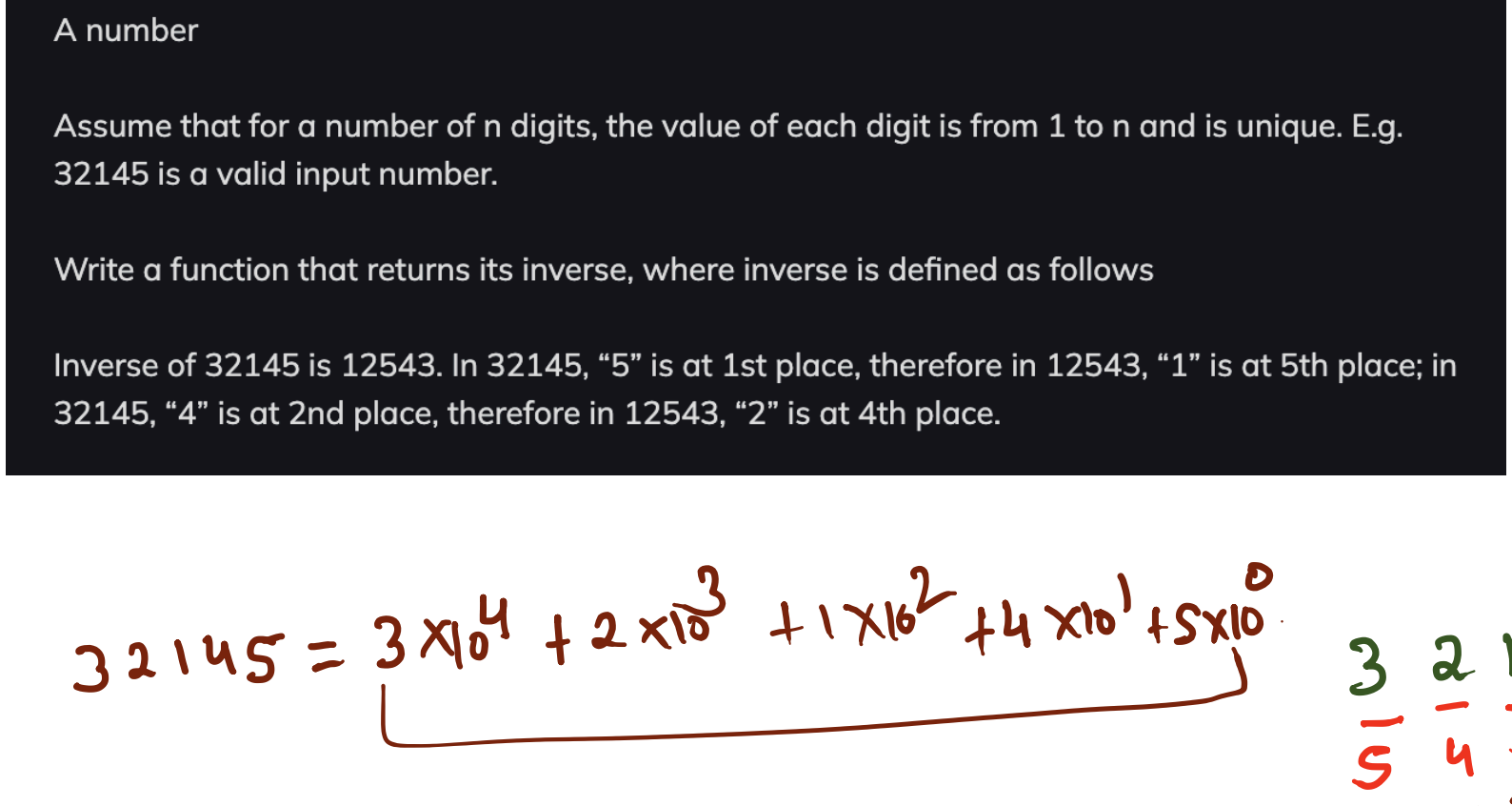

a=16 b=7
c=a-b=9
sub
a=7, b=9
c=16
ret c+sub(c,a)
16 9
25
16
addition
main
Syso(Hey)
a=7, b=9
Syso(addition(a,b))
Syso(okay)

```
static int val = 100;

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        int a = 7;
        int b = 9;
        System.out.println(val);//100
        System.out.println(addition(a, b));// 25
        System.out.println(val);
        System.out.println("Okay");

    }
    public static int addition(int a, int b) {
        val = val - 5;
        int c = a + b;
        return c + sub(c, a);
    }
    public static int sub(int a, int b) {
        return a - b;
    }
```


95
16
a=7, b=9
Val=Val-5
25
Hey
100
add
Syso(Hey)
a=7, b=9
Syso(Val)
Syso(addition(a,b))
Syso(val)
Syso(ok
main

```
static int val = 100;

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        System.out.println("Hey");
        int a = 7;
        int b = 9;
        System.out.println(val);// 100
        System.out.println(addition(a, b));// 25
        System.out.println(val);
        System.out.println("Okay");

    }
    public static int addition(int a, int b) {
        int val = 80;
        val = val - 5;
        int c = a + b;
        return c + sub(c, a);
    }
    public static int sub(int a, int b) {
        return a - b;
    }
```


Val=100
16
a=7, b=9
Val=80
Val=Val-5
75
Hey
100
add
main



Inverse of number

Take the following as input.

A number

Assume that for a number of n digits, the value of each digit is from 1 to n and is unique. E.g. 32145 is a valid input number.

Write a function that returns its inverse, where inverse is defined as follows

Inverse of 32145 is 12543. In 32145, “5” is at 1st place, therefore in 12543, “1” is at 5th place; in 32145, “4” is at 2nd place, therefore in 12543, “2” is at 4th place.

3 2 1 4 5
5 4 3 2 1

12543

1 2 5 4 3
5 4 3 2 1

$$32145 = \underbrace{3\times10^4 + 2\times10^3 + 1\times10^2 + 4\times10^1 + 5\times10^0}$$

3 2 1 4 5
5 4 3 2 1

3 2 1 4 5
5 4 3 2 1

place×10

$$1\times10^{5-1} + 2\times10^{4-1} + 3\times10^{1-1} + 4\times10^{2-1} + 5\times10^{3-1}$$

```
10000
 2000
    3
   40
  500
-----
12543
```

3 2 1 4 5
3 4 5 2 1

3214 → 321 → 32 → 3 → 0

2 3 4

$$1\times10^{5-1} + 2\times10^{4-1} + 3\times10^{1-1} + 4\times10^{2-1} + 5\times10^{3-1}$$

place = ~~1~~ ~~2~~ ~~3~~ 4 5

```
while (n>0){
    rem = n % 10
    sum = sum + place × 10^(rem-1)
    n = n/10
    place++
}
```

```
while(n>0){
    n=n/10
    c++
}
```

321 → 32 → 3 → 0

1634

1634 → 163 → 16 → 1 → 0

$4^4 + 3^4 + 6^4 + 1^4$

cod=4

$4^4 + 3^4 + 6^4 + 1^4$

```
while(n>0){
    rem=n%10
    sum=sum+rem^cod
    n=n/10
}
```

```
import java.util.Scanner;

public class Is_Armstrong_Number {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        System.out.println(Is_Armstrong(n));
    }

    public static boolean Is_Armstrong(int n) {
        int cod = countofdigit(n);
        int sum = 0;
        while (n > 0) {
            int rem = n % 10;
            sum = (int) (sum + Math.pow(rem, cod));
            n = n / 10;
        }
        if (sum == n) {
            return true;
        } else {
            return false;
        }
    }

    public static int countofdigit(int n) {
        int c = 0;
        while (n > 0) {
            n = n / 10;
            c++;
        }
        return c;
    }
}
```

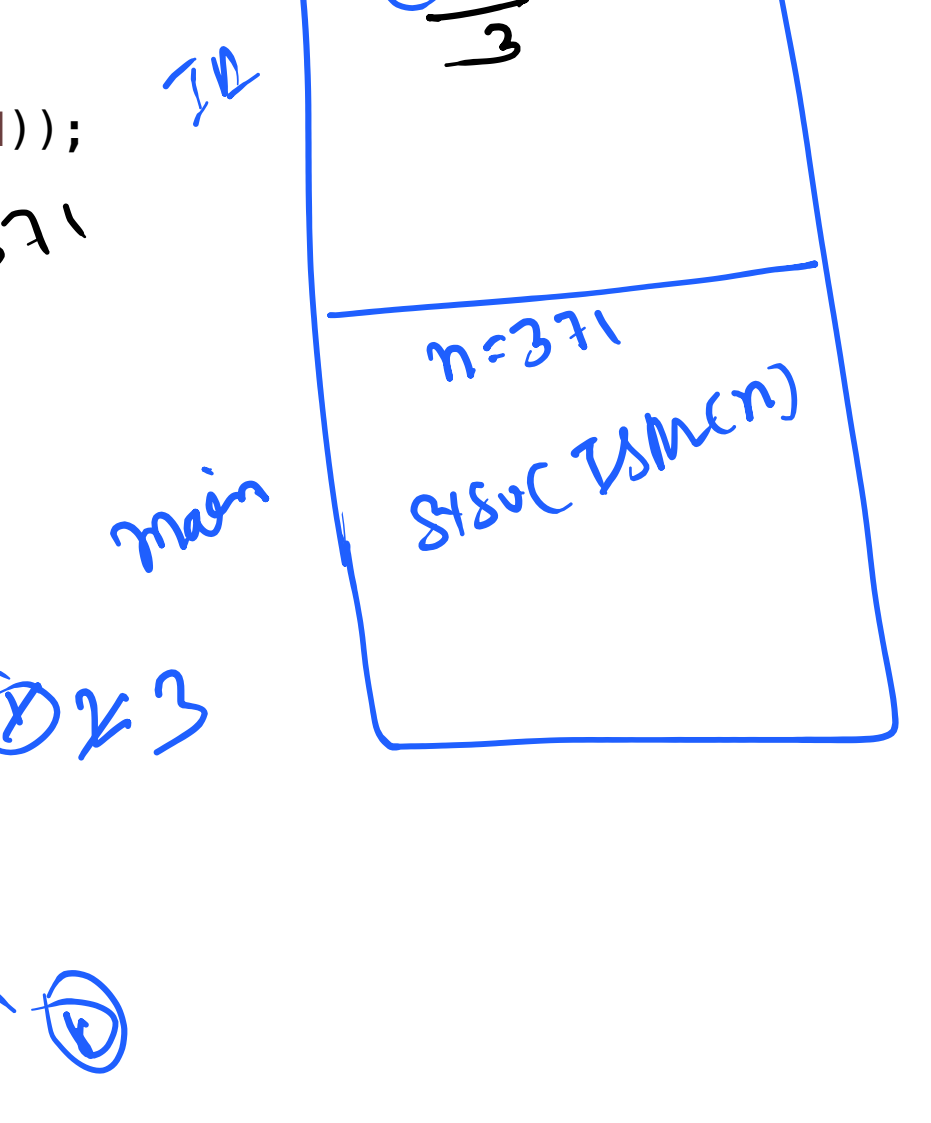
371 → 37 → 3 → 0

371 0

$3^3 + 7^3 + 1^3$ 371

371 > 0 → 37 → 3 → 0

① 2 3


n=371
count of digit
n=371
cod=c(371)
3
IA
n=371
Syso(ISA(n))
main